Fall, 2024

Marmara University
Engineering Faculty
Electrical- Electronics Engineering
EE4078- Fundamentals of Computer Networks
Second Midterm Exam Solutions

1.

(5) a) TCP slowstart is operating in the intervals [1,6] and [23,26]
(5) b) TCP congestion avoidance is operating in the intervals [6,16] and [17,22]
(5) c) After the 16th transmission round, packet loss is recognized by a triple duplicate ACK. If there was a timeout, the congestion window size would have dropped to 1.
(5) d) After the 22nd transmission round, segment loss is detected due to timeout, and hence the congestion window size is set to 1.
(5) e) The threshold is initially 32, since it is at this window size that slow start stops and congestion avoidance begins.

2.

Step	N'	D(t),p(t)	D(u),p(u)	D(v),p(v)	D(w),p(w)	D(y),p(y)	D(z),p(z)
0	x	∞	∞	3,x	6,x	6,x	8,x (1)
1	xv	7,v	6,v	3,x	6,x	6,x	8,x (4)
2	xvu	7,v	6,v	3,x	6,x	6,x	8,x (4)
3	xvuw	7,v	6,v	3,x	6,x	6,x	8,x (4)
4	xvuwy	7,v	6,v	3,x	6,x	6,x	8,x (4)
5	xvuwyt	7,v	6,v	3,x	6,x	6,x	8,x (4)
6	xvuwytz	7,v	6,v	3,x	6,x	6,x	8,x (4)

3. (a) In the case of IP and UDP, these are unreliable protocols that do not guarantee delivery, so they do not notify the source. TCP does guarantee delivery. However, the technique that is used is a timeout. If the source does not receive an acknowledgment to data within a given period of time, the source retransmits. (10) (5) (5)
(b) UDP has a fixed-sized header. The header in TCP is of variable length.

4. (a) Maximum number of hosts = $2^9 - 2 = 510$. The reason we must subtract 2 from the final number is because there are always 2 addresses allocated for each address block: the subnet ID (the first address) and the broadcast address (the last address). (5)

(b) Subnet A has 113 hosts, so it will need at least 115 addresses. The least number of bits that satisfy this is 7 bits. We take the prior subnet and add 128, the result of which is 215.176.218.0/25. (4)
The broadcast address of subnet A is 215.176.218.127 (the last address in the IP range). (2) The first IP address of subnet A is 215.176.218.1 (adding 1 to the subnet address). The last IP address of subnet A is 215.176.218.126 (subtracting 1 from the broadcast address). (2) (2)

(c) Subnet B has 62 hosts, so it will need at least 64 addresses. The least number of bits that satisfy this is 6 bits. We take the prior subnet and add 64, the result of which is 215.176.218.128/26. The broadcast address of subnet B is 215.176.218.191 (the last address in the IP range). (2) The first IP address of subnet B is 215.176.218.129 (adding 1 to the subnet address). (2) The last IP address of subnet B is 215.176.218.190 subtracting 1 from the broadcast address). (4) (2)